

Amy Munson

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Experience

FREELANCE

Full Stack Software Engineer

Saratoga, CA

December 2025 – Present

- Designed and built a full-stack real estate brokerage web app using Next.js, JavaScript, and PostgreSQL, featuring dynamic property listings, showing requests, and a secure admin dashboard for managing listings and inquiries.
- Overhauled the budgeting system of a TypeScript AI travel planning app and revamped the media ingestion and classification, notification, and social functionality and user interface.
- Built an admin dashboard that consolidates cost and usage metrics from external services and charts trends.
- Leveraged Claude Code throughout the development lifecycle to accelerate delivery and debugging.

CAHSI AND UNIVERSITY OF CALIFORNIA, SANTA CRUZ

Santa Cruz, CA

Research Scholar

February 2024 – May 2025

- Conducted research on the use of large language models in identifying code vulnerabilities in comparison to traditional static analysis tools with Professor Alvaro Cardenas and AppSecAI.
- Defined custom prompts to demonstrate the effects of prompt styles on LLM performance.
- Analyzed data using metrics such as accuracy, F1-score, and AUC to evaluate LLM performance against static analysis, compare results from multiple models, and assess performance across different CWEs.
- Co-authored a paper accepted and presented at the IEEE/ACM International Conference on Software Engineering 2025 on large language models in vulnerability detection: <https://ieeexplore.ieee.org/document/11028575>

UNIVERSITY OF CALIFORNIA, SANTA CRUZ

Santa Cruz, CA

Tutor and Grader

October 2023 – March 2024

- Led in-person tutoring sessions for Computer Systems and C Programming to clarify key course concepts and provide individualized support in debugging C code and virtual machine environments.
- Graded coding assignments for a class of 200 students and delivered feedback on student reports, focusing on depth of analysis and understanding of key concepts.

Education

UNIVERSITY OF CALIFORNIA, SAN DIEGO

La Jolla, CA

Master of Science in Computer Science

December 2025

- GPA: 3.73
- Relevant Coursework: Computer Security, Networked Systems, Database Systems, Operating Systems Principles, Parallel Computation, Principles of Software Engineering

UNIVERSITY OF CALIFORNIA, SANTA CRUZ

Santa Cruz, CA

Bachelor of Science in Computer Science

June 2024

- GPA: 3.88
- Relevant Coursework: Distributed Systems, Web Applications, Natural Language Processing, Computer Security, Data Structures and Algorithms, Technical Writing

Skills and Projects

Languages: Python, C, C++, Golang, Java, SQL, JavaScript, TypeScript, HTML, CSS, Bash, Git, Linux

Tools: Docker, React, Vue, Node.js, Next.js, Flask, AWS, CUDA, TensorFlow, PyTorch, Pandas, Claude Code

LLM Agent with Access Control Research Project

- Developed a secure function-calling AI Agent that predicts required tools for each user prompt and dynamically enforces tool access policies specific to the task requirements.
- Designed a reference-monitor that validates all tool calls against the XML prediction of required tools to prevent unauthorized access and prompt injection attacks.
- Implemented XML prediction generation with OpenAI and Llama models, integrated third-party APIs for agent tools, and developed an evaluation framework with a test suite for prompt injection testing.

Publications

A. Munson, J. Gomez and A. A. Cardenas, "♪ With a Little Help from My (LLM) Friends: Enhancing Static Analysis with LLMs to Detect Software Vulnerabilities," in 2025 IEEE/ACM International Workshop on Large Language Models for Code (LLM4Code), Ottawa, ON, Canada, 2025, pp. 25-32, doi: 10.1109/LLM4Code66737.2025.00008.